

ASSESSMENT-2

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Course: Computer Vision With OpenCV

Code: DSA0202

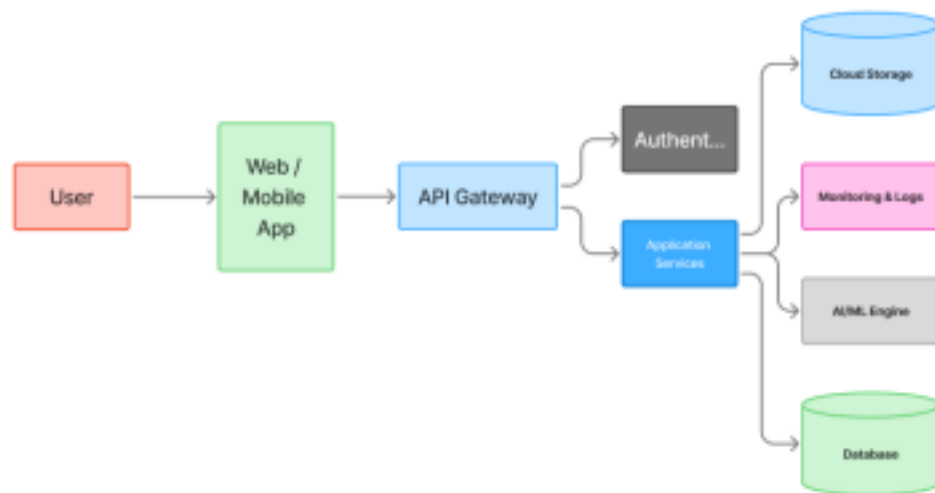
Faculty: Dr Mayuranathan M

Question 1: Smart Photo Editing + Object Recognition + AR Application

The system captures an image and applies image processing techniques to improve its quality. Computer vision detects and recognizes objects using different levels of vision. The recognized objects are then enhanced with computer graphics to generate AR effects. Finally, the application provides features such as smart photo editing, object recognition, and augmented reality experiences.

Key Points

- Image acquisition and enhancement
- Object detection and recognition
- Scene understanding
- AR visualization using computer graphics
 - Real-world applications: Snapchat filters, Pokémon GO, Google Lens



Question 2: Autonomous Drone for 3D Scene Reconstruction

The drone captures multiple images using onboard cameras. Camera calibration removes lens distortion and estimates camera parameters. Binocular vision and multiple-view geometry calculate depth information from different viewpoints. Finally, the system reconstructs an accurate 3D model of the environment for mapping and navigation.

Key Points

- Image capture by drone
- Camera model and image formation
- Camera calibration
- Depth estimation using binocular vision
- 3D scene reconstruction
- Applications: Drone mapping, autonomous navigation, digital twins

